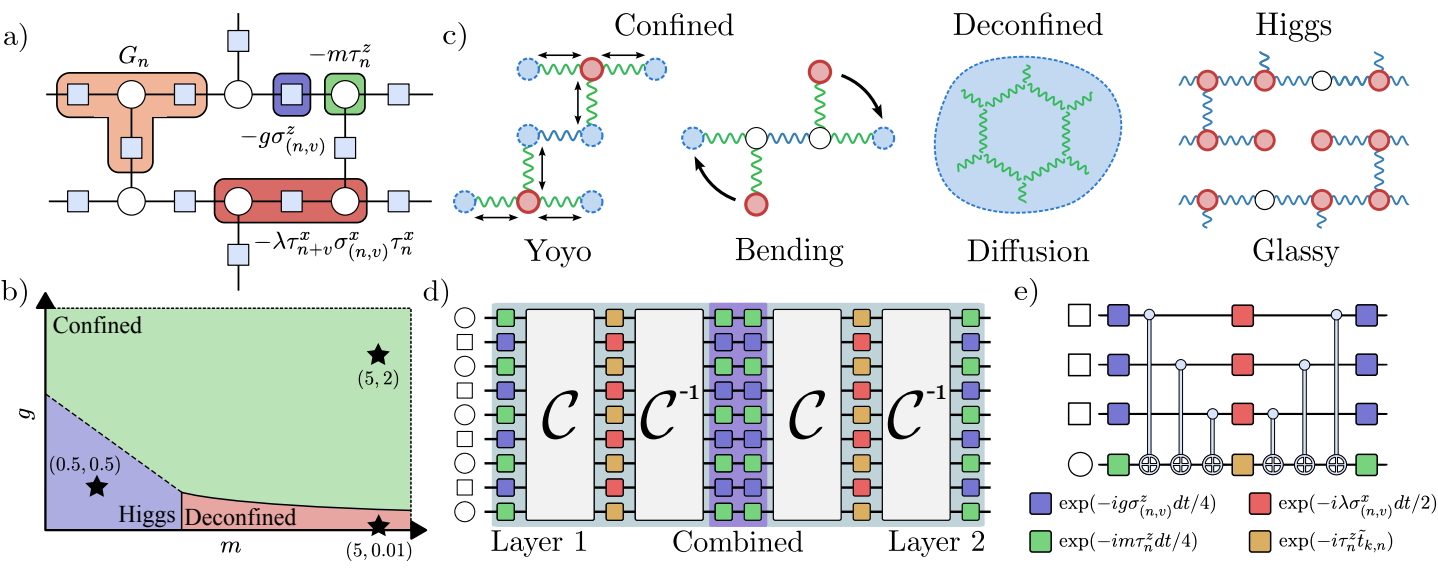

a)
G_n
$-m\tau_n^z$
$-g\sigma_{(n,v)}^z$
$-\lambda\tau_{n+v}^x\sigma_{(n,v)}^x\tau_n^x$
b)
Confined
(5, 2)
(0.5, 0.5)
Higgs
Deconfined
(5, 0.01)
g
m
c)
Confined
Deconfined
Higgs
Yoyo
Bending
Diffusion
Glassy
d)
$\mathcal{C}$
$\mathcal{C}^{-1}$
$\mathcal{C}$
$\mathcal{C}^{-1}$
Layer 1
Combined
Layer 2
e)
$\exp(-ig\sigma_{(n,v)}^z dt/4)$
$\exp(-i\lambda\sigma_{(n,v)}^x dt/2)$
$\exp(-im\tau_n^z dt/4)$
$\exp(-i\tau_n^z \tilde{t}_{k,n})$